Answer C

3) Car travels at constant acceleration, hence using $s = \frac{1}{2} a t^2$, distance travelled between 1 s and 2 s, is equal to

$\frac{1}{2} a (2)^2 - \frac{1}{2} a (1)^2 = 12$
Hence, $a = 8$

Hence, for distance travelled between t =3s and t = 4s,
distance travelled = $\frac{1}{2} a (4)^2 - \frac{1}{2} a (3)^2 = 28$ m
Answer A

4) Using conservation of energy, the speed of the stone being thrown up will be equal to